

A-Level Combination Guide — Andrey

Comparing the two candidate combinations · STEM program access · Bioinformatics deep-dive

■ United States (primary focus)

No subject-specific prerequisites. All biology and chemistry taught from scratch. Maths+FM+Physics+Chemistry blocks **nothing**. MIT, Stanford, UCSD, CMU, Harvard — all fully accessible.

■ France — same conclusion as US

A-levels accepted as Baccalauréat equivalent. No Biology required for STEM Licences. **Paris-Saclay (ARWU #15)** Bioinformatics is a Master's entered after Maths/CS Licence. Main barrier: French language at UG level.

■ United Kingdom — different picture

Subject-specific gates: Biology A-level treated as prior knowledge. Edinburgh/Manchester/Imperial have no UG Bioinformatics. UCL and Cambridge require Biology. UK is the outlier.

■ Chinese universities: to be checked separately — expected finding similar to US/France.

RECOMMENDATION SUMMARY

■ Choose Maths + FM + Physics + Chemistry if targeting US / France / China

From a US/France university perspective, this combination blocks almost nothing. Top US programs (MIT, Stanford, UCSD, CMU) and Paris-Saclay teach all biology and biochemistry from scratch — no prior Biology A-level is expected or required.

Maths+FM+Physics+Chemistry is an exceptionally strong profile: Chemistry is directly useful since all Bioinformatics programs include General Chemistry and Biochemistry in years 1–2. Bioinformatics, Computational Biology, CS, Engineering, and Physical Sciences are all fully open.

Recommended choice if Andrey is targeting US, French, or Chinese top universities.

■ Biology only matters if UK universities are the primary target

In the UK system, Biology A-level is required for UCL Computational Biology, Cambridge Biological NatSci, and most Biochemistry/Genetics degrees. Edinburgh, Manchester, and Imperial don't even offer UG Bioinformatics — it's postgraduate only there. If Andrey is primarily targeting UK top universities for biology-adjacent programs, Biology is needed. For US, China, or France: Chemistry wins every time.

SECTION 1 — THE TWO COMBINATIONS

Option A — Physics & Chemistry Track

Maths · Further Maths · Physics · **Chemistry**

The hardest all-round STEM combination. All four are facilitating subjects. **For US universities: opens every STEM program including Bioinformatics and Computational Biology** — biology is taught from scratch. Chemistry keeps medicine on the table globally. **Recommended for US-targeting students in any quantitative direction.**

Option B — Physics & Biology Track

Maths · Further Maths · Physics · **Biology**

Also strong, but more specialised toward UK life sciences. Opens UK-specific Biology-gated programs. For US universities, Biology offers no meaningful advantage over Chemistry — US programs teach all biology from scratch. Ironically **weakens medicine applications** globally since most med schools require Chemistry, not Biology. Best only if Andrey targets UK life science programs specifically.

SECTION 2 — WHICH PROGRAMS DOES EACH OPEN? (US UNIVERSITY PERSPECTIVE)

■ US context: US universities do not require specific high school subjects — they teach all biology, chemistry, and physics from scratch. The "blocked" column below contains UK-only restrictions that do not apply to US universities.

✓ Open with **EITHER** combination (US universities)

■ UK-only restrictions (not applicable to US)

● Mathematics / Applied Mathematics	Maths + FM is all that's needed anywhere in the world
● Statistics & Actuarial Science	Maths-driven; no biology or chemistry required
● Computer Science	Maths is the primary requirement at all top universities
● Data Science / AI	Maths + FM more than sufficient
● Physics / Astrophysics	Maths + Physics; 4th subject irrelevant
● All Engineering branches	Mechanical, Electrical, Aerospace, Civil, Biomedical
● Earth & Environmental Sciences	Chemistry or Physics usually sufficient
● Materials Science / Chemical Engineering	Chemistry + Physics + Maths is the ideal combo
● Bioinformatics / Computational Biology	MIT, Stanford, UCSD, CMU — all teach biology from scratch
● Biochemistry / Molecular Biology (US)	US universities teach all prerequisites at university level
● Neuroscience (US)	US programs have no high school Biology requirement

These restrictions apply in the UK system only . From a US admissions standpoint, choosing Chemistry over Biology does not block any of these programs.	
● Biochemistry (UK top unis)	Oxford, Cambridge, Imperial require Chemistry + Biology. UK-specific gate.
● Genetics / Molecular Biology (UK)	Biology A-level required at virtually all UK universities
● Zoology / Ecology (UK)	Pure biology degrees; Biology A-level essential in UK
● Neuroscience (UK top unis)	UCL, Edinburgh prefer or require Biology at UK level
● Bioinformatics UG at top UK unis	Edinburgh/Manchester/Imperial = postgrad only. UCL/Cambridge require Biology.

■ Green = fully open at US top universities regardless of Chemistry vs Biology. ■ Yellow = partly affected in UK. ■ Red = blocked in UK but NOT in US.

SECTION 3 — PROGRAM RESTRICTIONS BY UNIVERSITY SYSTEM		
■ US Universities — Nothing is blocked At MIT, Stanford, UCSD, CMU, Harvard: choosing Chemistry over Biology does not block any STEM program. The university teaches everything from scratch in year 1. A student with Maths+FM+Physics+Chemistry is actually ideally prepared. ■ UK Universities — Biology A-level is a real gate (for reference only) The UK operates a subject-specific admissions system. Choosing Chemistry without Biology does close certain programs at top UK universities. This section is provided for completeness — if UK is not Andrey's target, it is not relevant.		
■ UK-only restrictions when choosing Chemistry instead of Biology ■ The following restrictions apply in the UK system only. They do NOT apply to US, French, or Chinese universities. Biochemistry (UK) Oxford, Cambridge, Imperial, UCL, Edinburgh all require both Chemistry and Biology. Chemistry alone is not sufficient. UK: Fully blocked		

Genetics (UK)	Biology A-level is a universal UK requirement. DNA and cellular mechanisms taught at A-level are assumed from day one.	UK: Fully blocked
Computational Biology (UK)	UK programs expect Biology as entry knowledge. Not applicable in the US where CMU, Stanford, MIT teach biology from scratch.	UK: Fully blocked
Neuroscience (UK top unis)	UCL, Edinburgh, Manchester prefer or require Biology A-level. Some UK universities accept Chemistry as alternative.	UK: Mostly blocked
Bioinformatics UG (UK top unis)	Edinburgh = MSc only. Manchester = MSc only. Imperial = MRes only. UCL requires Biology at A*. Cambridge Biological NatSci = 98% take Biology.	UK: Blocked at top unis
Bottom line: If Andrey targets US universities, Maths+FM+Physics+Chemistry blocks nothing relevant to Bioinformatics or Computational Biology. The restrictions above are specific to the UK admissions system and do not transfer to other countries.		

SECTION 6 — ■ PRIMARY REFERENCE: TOP US UNIVERSITIES (ARWU)

US universities are the primary frame of reference for this analysis

At top US universities, admission is based on overall academic strength — not on specific subject combinations. All prerequisite biology, chemistry, and physics is taught from scratch within the degree. Maths + FM + Physics + Chemistry is a strong profile for every program below.

University	UG Bioinformatics / CompBio degree	Biology prior req?	Open?
MIT (ARWU #3)	Course 6-7: Computer Science & Molecular Biology — joint BS from EECS + Biology. Biology, genetics, biochemistry, and CS algorithms for genomics. Biology taught from scratch.	No — biology taught from scratch within the degree.	Yes
Stanford (ARWU #2)	BS Biomedical Computation (School of Engineering). Foundational Math, Biology, Chemistry, Physics, CS all taken at university in years 1–2. Four tracks. Mandatory research.	No — all foundational science taught as part of the degree in years 1–2.	Yes
UC San Diego (ARWU #14)	BS Bioinformatics (BI34) — one of few true standalone UG Bioinformatics degrees worldwide. Biology (BILD 1,3,4,5), Chemistry, Organic Chem, Maths, Physics, CS all first-year university courses.	No — all biology and chemistry prerequisites are first-year university courses.	Yes
Carnegie Mellon (ARWU ~25)	BS Computational Biology (School of Computer Science). Biological Core: Honors Modern Biology, Genomics, Biochemistry, Cell Biology — all taught within degree. Guaranteed research.	No — all biological core taught within degree in years 1–3.	Yes
Harvard (ARWU #1)	No standalone Bioinformatics BS. Applied Mathematics or CS concentrations with bioinformatics content. Harvard's flexible concentration system allows substantial bioinformatics without a dedicated degree.	Not required — strong maths/CS is the primary asset.	Yes

■ **Key advantage of Maths+FM+Physics+Chem for US programs**
US Bioinformatics programs teach all biology from scratch. Chemistry A-level is directly useful — these programs include General Chemistry and Biochemistry in years 1–2. A student with strong Maths, Physics, and Chemistry is ideally prepared.

■ ■ **What US universities DO look for**
Strong GPA, SAT/ACT or international equivalent (A-levels), evidence of intellectual curiosity. For competitive programs like MIT 6-7 and Stanford BMC, extracurricular research, coding projects, or science competitions matter more than which specific science A-level was taken.

SECTION 4 — BIOINFORMATICS VS COMPUTATIONAL BIOLOGY

Bioinformatics		Computational Biology
How do we store, process, and extract meaning from biological data at scale?	Core question	How do we build mathematical models that explain how biological systems behave?
Python, R, databases, machine learning, sequence alignment algorithms	Primary tools	Differential equations, stochastic modelling, physics-based simulation, R/Python
Analysing genome sequences, building variant-calling pipelines, training ML models on protein data	Typical work	Modelling protein folding, simulating cell signalling networks, evolutionary dynamics
Usually Computing, Informatics, or joint Schools	Faculty home	Usually Life Sciences, Biology, or Biosciences
Maths/CS background; Biology taught from scratch	Entry profile	Biology/chemistry background; maths taught alongside
US: Maths + strong academics — no prior Biology needed. UK: Edinburgh/Manchester/Imperial have no UG program; UCL requires Biology.	A-level req.	US: No specific requirement — Biology taught within degree. UK: Biology expected; Chemistry alone often insufficient.
Students who are data/CS-first and want to apply skills to medicine and genomics	Best for	Students who are biology-first and want to bring rigour and modelling into their science

■ **Bioinformatics is currently in significantly higher demand**

This is not a close call. The genomics revolution, AI drug discovery, and NHS digitalisation have all created acute demand for people who can handle biological data at scale — exactly the Bioinformatics skill set. Computational Biology, while intellectually deep, produces graduates aimed mainly at academic research, which is a narrower pipeline.

~£35–45k UK Bioinformatics graduate starting salary (vs ~£28–32k Computational Biology)	3–5× More industry job postings for Bioinformatics vs Computational Biology roles	\$1.8T Projected global genomics market by 2030 — the primary employer of Bioinformatics graduates
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Key employers: Illumina · AstraZeneca · GSK · DeepMind Health · Recursion Pharmaceuticals · Wellcome Sanger Institute · NHS Genomic Medicine Service · Oxford Nanopore · dozens of VC-backed biotech startups.

SECTION 7 — ■■ FRANCE: PARIS-SACLAY UNIVERSITY (VERIFIED)

■■ **What is Paris-Saclay?**

ARWU #15 globally — significantly higher-ranked than most UK universities. Paris-Saclay is a confederation including CentraleSupélec, ENS Paris-Saclay, AgroParisTech, Institut d'Optique, and several grandes écoles. For STEM it is world-class, particularly in mathematics, physics, and computational sciences.

■ **A-level equivalence: Chemistry combo is fine**

France does not gatekeep university programs by specific high school subjects. British A-levels are accepted as equivalent to the French Baccalauréat. For STEM Licences, **Maths A-level is the key signal** — Chemistry is a positive, Biology is not required at any stage. The French system assesses overall academic profile. Maths+FM+Physics+Chemistry is an excellent profile.

Program / Route	Structure	Biology req?	Language	Open ?
Licence Informatique (CS Bachelor, 3 years)	Primary feeder into Master Bioinformatics for CS-background students.	No — Maths is the core requirement.	French	■ Yes
Licence Mathématiques (Maths Bachelor, 3 years)	Strong feeder into M1 BIBS-IA Bioinformatics track (accepts pure Maths backgrounds with remedial bridging).	No — pure mathematics track.	French	■ Yes
Master Bioinformatics M1 BIBS-IA (after Licence, 1 year)	Accepts students from single-discipline backgrounds: Biology alone, Maths alone, OR CS alone. Built-in remedial bridging makes it accessible to students with no prior biology.	No — biology taught within the Master. Maths or CS background alone accepted.	French	■ Yes
Master Bioinformatics M1 GENIOMHE-AI (after Licence, 1 year)	Prefers multidisciplinary profile (Maths/CS + some biology). Taught in English — M2 covers Genomics, AI, Mathematical modelling.	Preferred but not mandatory — strong Maths alone still acceptable.	English	■■ Partial
CentraleSupélec International Class (engineering entry, 1 prep year)	Dedicated pathway for non-French students into Paris-Saclay engineering via CentraleSupélec. Strong quantitative profile required.	No — engineering/maths/physics focused.	English/French	■ Yes

■ Bottom line: same picture as US

Maths+FM+Physics+Chemistry blocks nothing in the Paris-Saclay Bioinformatics pathway. The route is Licence (Maths or CS, 3 years) → Master Bioinformatics M1 BIBS-IA. Biology taught from scratch. No specific A-level subject gates.

■■ Practical consideration: French language

Most UG Licences are taught in French (B2–C1 required). The GENIOMHE-AI Master's track and CentraleSupélec International Class are in English. French language proficiency is the real barrier, not A-level subject choice.

■■ Chinese universities (Tsinghua, Peking, Fudan, SJTU, Zhejiang) — to be verified separately. Expected finding: similar to US/France — no specific high school biology prerequisite for STEM programs; Maths+Physics+Chemistry A-level profile is well-regarded.